



Syrphid fly response to urban heat islands varies with functional traits

Frédéric McCune¹ · Étienne Normandin² · Amélie Gervais¹ · Marc J. Mazerolle³ · Valérie Fournier¹

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Abstract

Syrphid flies (Diptera, Syrphidae) play important ecological roles as pollinators, pest control agents, and decomposers. Communities are influenced by environmental variables such as land use. However, these interactions are poorly studied in urban landscapes. We used pan traps in community gardens, cemeteries, and parks in Montreal, Canada, to investigate syrphid fly communities in an urban landscape and describe how site surface area and heat island cover shape these communities. We estimated the species richness of communities and the occupancy patterns of three syrphid functional groups (insectivores, terrestrial saprophagous, and aquatic saprophagous). We captured 1,791 specimens from 48 species. Species richness increased with site surface area but did not vary with the level of urbanisation. The occupancy of the three groups did not change with site surface area. Syrphid flies of the terrestrial saprophagous group were less likely to occur on sites with high urbanisation, but the occupancy of insectivore and aquatic saprophagous groups did not vary with the variable.

Implications for insect conservation Our results highlight that cities can harbour diversified syrphid fly communities and that certain functional groups are tolerant to urbanisation pressures. Our study suggests that conservation measures targeting syrphid flies in cities must be adapted to functional traits and larval requirements.

Keywords Hoverflies · Flower flies · *Toxomerus* · Hierarchical models · Urban warming · Impervious surfaces

Introduction

Syrphid flies (Syrphidae) form one of the largest families of Diptera, with more than 6,000 species described worldwide (Courtney et al. 2017). However, many of their ecological roles have been underestimated (Orford et al. 2015; Rader et al. 2016; Rodriguez-Gasol et al. 2020). Syrphid flies play three crucial ecological roles in ecosystems. Specifically, larvae contribute to pest control and decomposition of organic matter, whereas adults are important pollinators (Doyle et al. 2020; Dunn et al. 2020; Rader et al. 2016; Ssymank et al. 2008; Wotton et al. 2019). In fact, syrphids are the most frequent non-bee family visitors to flowers of crops (Rader et al. 2020).

The feeding habits of syrphid flies vary greatly between species and life stages. Nearly all species feed on pollen and nectar at the adult stage (Sommaggio 1999). Pollen is mainly used by females for the maturation of ovaries, whereas nectar is consumed by both males and females for energy (Hasslett 1989). Unlike bees, syrphid flies are not central place foragers and do not commute to and from a nest to feed their offspring. As such, adult syrphids can transport pollen over longer distances than bees (Doyle et al. 2020). In contrast

✉ Frédéric McCune
frederic.mccune.1@ulaval.ca
Étienne Normandin
etienne.normandin@gmail.com
Amélie Gervais
amelie.gervais.bio@gmail.com
Marc J. Mazerolle
marc.mazerolle@sbf.ulaval.ca
Valérie Fournier
Valerie.Fournier@fsaa.ulaval.ca

¹ Centre de recherche et d'innovation sur les végétaux, Département de phytologie, Faculté des sciences de l'agriculture et de l'alimentation, Université Laval, 2480, Boul. Hochelaga, Québec, Qc G1V 0A6, Canada
² Centre sur la biodiversité, Département des sciences biologiques, Université de Montréal, Montréal, Qc H1X 2B2, Canada
³ Centre d'étude de la forêt, Département des sciences du bois et de la forêt, Faculté de foresterie, de géographie et de géomatique, Université Laval, Québec, Qc G1V 0A6, Canada

to adult syrphids, larval feeding strategies are diverse and include zoophagy, saprophagy, phytophagy, and mycophagy (Owen 1981; Rotheray and Gilbert 2011; Sommaggio 1999). This array of feeding strategies promotes the provision of a wide variety of ecosystem services by syrphid larvae. For instance, aphidophagous syrphid larvae contribute to pest control and can be reared as commercial biocontrol agents (Dunn et al. 2020; Moerkens et al. 2021; Pekas et al. 2020), whereas saprophagous larvae decompose a considerable amount of organic matter (Moquet et al. 2018; Owen 1981; Rotheray and Gilbert 2011).

The habitat requirements of syrphid fly communities are as diverse as their life history traits and must be considered at both landscape and local scales (Meyer et al. 2009; Moquet et al. 2018). Adults mostly need flower-rich open habitats. In contrast, larvae live in habitats such as aphid-infested agricultural fields, forests, and ponds. Thus, syrphid communities are positively influenced by landscape diversity (Meyer et al. 2009; Sommaggio 1999). At the local scale, factors shaping syrphid fly communities include floral density and diversity, as well as patch size (Grossmann et al. 2022; Meyer et al. 2009; Moquet et al. 2018). Increasing patch size is linked to higher resource abundance and diversity (Ouin et al. 2006) and therefore results in positive impacts on syrphid flies, including greater community richness (Herrault et al. 2016; Jauker et al. 2019; Meyer et al. 2009). The importance of such local-scale factors for the occurrence of syrphid species depends on their life history traits, such as larval diet and microhabitat, as well as adult body size (Dziock 2006; Meyer et al. 2009; Moquet et al. 2018; Speight and Castella 2001).

Urbanisation influences habitats and plant and animal communities (Deguines et al. 2016; Knop 2016; McKinney 2006), including plant-pollinator networks (Fisogni et al. 2022; Rivkin et al. 2020). However, only a few European studies have investigated the impact of urbanised landscapes on syrphid fly communities. They found that the abundance and richness of syrphid communities are negatively impacted by urbanisation (Baldock et al. 2015; Bates et al. 2011; Grossmann et al. 2022; Luder et al. 2018; Persson et al. 2020; Udy et al. 2020; Verboven et al. 2014) and that syrphid fly assemblages in cities can be a subset of those in rural areas (Persson et al. 2020). However, these studies are geographically restricted and only superficially investigated many aspects of urbanisation. Indeed, urban landscapes are also characterised by warmer temperatures and the presence of heat islands (OECD 2010), but the impact of urban heat islands on syrphid fly communities has not been studied to date. Heat islands are caused by the prevalence of impervious surfaces and the loss of vegetation (Hart and Sailor 2009; Jenerette et al. 2011), and can increase temperatures by up to 12 °C (Angilletta et al. 2007; Voogt et al. 2002). In fact, heat islands can be used as a proxy for urbanisation

(Hart and Sailor 2009; Jenerette et al. 2011; McCune et al. 2020). Higher temperatures can impact insect phenology (Harrison et al. 2015), reproduction (Chick et al. 2019), development (Kaiser et al. 2016), fertility (Sales et al. 2018), and heat tolerance (Angilletta et al. 2007). Noel et al. (2022) showed that increased temperatures can alter the development and survival rate of larvae of the aphidophagous syrphid *Episyrphus balteatus*. For instance, heat islands can reduce wild bee abundance regardless of their life history traits (Hamblin et al. 2018). Urban warming can also induce a phenological mismatch between pollinators and flowering plants, thereby impacting their mutualistic interactions (Fisogni et al. 2020).

The objectives of this study were to determine the composition of syrphid fly communities in the Montreal metropolitan area (Canada) and to quantify the impacts of site surface area and heat islands used as a proxy for urbanisation on syrphid fly communities. We analyzed the species richness as well as the occupancy of syrphid fly species at the functional group level. We hypothesised that syrphid flies respond positively to site surface area, but negatively to urban heat islands in the landscape.

Methods

Study sites and syrphid fly sampling

This study was conducted in the Montreal metropolitan area (Fig. 1). With a population of 4.1 million people, it is the second most populous urban center in Canada. It is spread over an area of 4604 km² and has a population density of 890.2 people per km² (Ville de Montréal 2018). Syrphid flies were sampled on a total of 25 sites in 2012 and 2013, specifically five parks, nine cemeteries, and 11 community gardens. Community gardens are a type of garden with clearly defined plots where citizens are responsible for cultivating and harvesting their plot. These gardens are used to grow herbs, vegetables, and fruits, but also ornamental flowers. The sites and methods are described in detail in McCune et al. (2020) and Normandin et al. (2017). The sites thereafter refer to the sampled parks, cemeteries, and gardens. We summarize the methods in the following paragraphs.

Syrphid fly communities were surveyed on all 25 sites in 2012 and 2013 using pan traps consisting of white, yellow, and blue standard 400-ml pan traps filled with soapy water (Normandin et al. 2017). Bowls were left white, or painted yellow or fluorescent blue (Krylon® paint), such that each colour was represented equally across the total number of bowls on each site. Bowls were individually placed on 60-cm wooden sticks and deployed 1-m apart along linear transects on the margins of flower beds, gardens, and natural vegetation. The number of pan traps varied with site

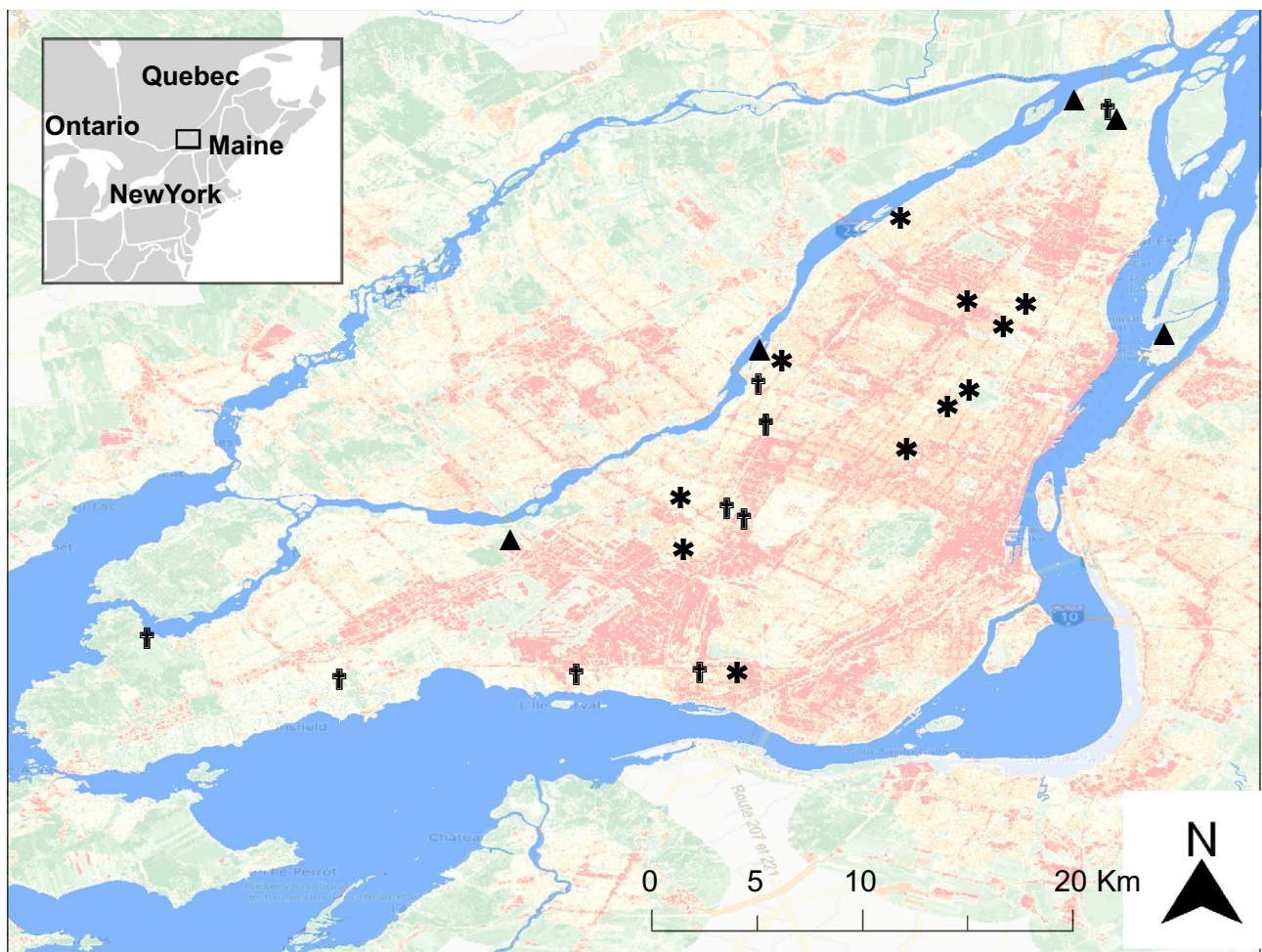


Fig. 1 Map of the study sites in the Montreal metropolitan area, QC, Canada. Cemeteries are represented by crosses, gardens by asterisks, and parks by triangles. The map background shows the heat island

mapping of Boulfroy et al. (2013) and Institut national de santé publique du Québec (INSPQ) (2013). The level of predicted heat islands ranges from green (lowest) to dark orange and red (highest levels)

surface area (three traps [i.e., one of each colour] for each 1,000 m² of site surface area), with a maximum of 45 traps on a given site. Sampling effort was expressed as the number of traps deployed during a given period and varied from 3 to 45 (mean of 20). From May to September, pan traps were deployed on each site about twice a month, for 48 h, for a total of six sampling periods in 2012 and seven periods in 2013.

Captured specimens were processed and pinned for identification to the lowest taxonomic level using keys and guides available for our study area (Miranda et al. 2013; Skevington et al. 2019). All specimens were identified to the species level except members of the genus *Eumerus*, *Pipiza*, *Platycheirus*, and *Sphaerophoria*, which are difficult to identify using external morphology (Skevington et al. 2019). Nonetheless, some male specimens of the genus *Sphaerophoria* could be identified to the species

level (*S. contigua* Macquart, 1847, *S. philanthus* Meigen and *S. pyrrhina* Bigot, 1884). Because most specimens of the *Sphaerophoria* genus could not be identified, they were grouped to the genus level for subsequent analyses. Voucher specimens are preserved at the laboratory of V. Fournier at Université Laval (Quebec, Canada). Species were characterised according to functional groups based on larval requirements. Larval functional traits were determined according to the habitat and nutritional needs of every syrphid species. It was based on published data on the syrphid flies of the study area (Skevington et al. 2019; Sommaggio 1999). Namely, we categorized species in four functional groups: zoophagous larvae (insectivores), terrestrial saprophagous or coprophagous larvae (terrestrial saprophagous), aquatic or semiaquatic saprophagous larvae (aquatic saprophagous), and phytophagous larvae feeding on living plants in terrestrial ecosystems (phytophagous).

Urban heat islands and site surface area

We extracted the data on the cover of urban heat islands from spatial databases (Boulfroy et al. 2013; Institut national de santé publique du Québec (INSPQ) 2013) and that of site surface area from Normandin et al. (2017). Site surface area refers to the surface area of each site (m^2) and varied from 567 to 147,000 m^2 (average of 19,980.32 m^2). We computed the cover of heat islands within 1 km around each sampling site as a proxy for urbanisation at the landscape level. We used the buffer tool in ArcGIS software (ArcMAP 10.5) to create buffers of 1 km around each site. Considering the dispersal ability of syrphid flies, we considered 1 km as a suitable distance to study the environmental variables that can influence syrphid fly communities at the landscape scale (Bates et al. 2011; Kleijn and van Langevelde 2006; Meyer et al. 2009). We calculated the proportion of heat island cover in each buffer zone using a public raster mapping of predicted urban heat islands and cool areas for the Montreal metropolitan area (Boulfroy et al. 2013; Institut national de santé publique du Québec (INSPQ) 2013). This mapping of heat islands was based on different variables including climatic data (surface temperature on multiple days and wind speed), an index of surface imperviousness, the normalized difference vegetation index (NDVI), and a land-use classification. Thus, the proportion of heat island cover reflected information from different variables associated with the degree of urbanisation at the landscape scale. Indeed, the effects of temperature and impervious surfaces are the key features of urbanisation which are expected to impact insect assemblages (Baldock 2020; Hamblin et al. 2018; Levé et al. 2019). The mapping of heat islands uses nine levels of increasing intensity, with levels eight and nine corresponding to heat islands (Boulfroy et al. 2013; Institut national de santé publique du Québec (INSPQ) 2013). Thus, we summed the area for these two levels in each buffer zone of 1,000 m-radius around each site to calculate the

proportion of the surface cover of heat islands in each buffer (Fig. 2).

Statistical analysis

To evaluate our sampling effort, we computed species accumulation curves using the vegan package (Oksanen et al. 2022). We pooled data for the three trap colours for each site and sampling period.

The probability of detecting species and individuals is rarely perfect or constant for a wide range of animal and plant taxa, as detection probability can vary due to several factors, including species behavior, sampling method, and habitat (MacKenzie et al. 2006; Williams et al. 2002). A few design modifications enable accounting for imperfect detection probability when estimating state variables such as species richness or occupancy from field studies (Kéry and Royle 2016; Mazerolle et al. 2007; Williams et al. 2002). Specifically, detection data collected in traps of three trap colors provided repeated observations to estimate species richness and occupancy of syrphid flies using a Bayesian framework. All numeric variables were standardized to zero mean and unit variance before analysis.

Species richness

We estimated the species richness of syrphids on each site during each trapping period using a hierarchical model based on data augmentation (Kéry and Royle 2016; Royle and Dorazio 2012; Royle et al. 2007). Data augmentation is an approach used to estimate the number of species in a community among an arbitrarily large set of M potential species (Royle and Dorazio 2012). This model was built using the observed detection histories of each species and incorporated the effects of explanatory variables on species richness and detection probability. Specifically, we translated capture data for each species into detected (1) or not detected (0)

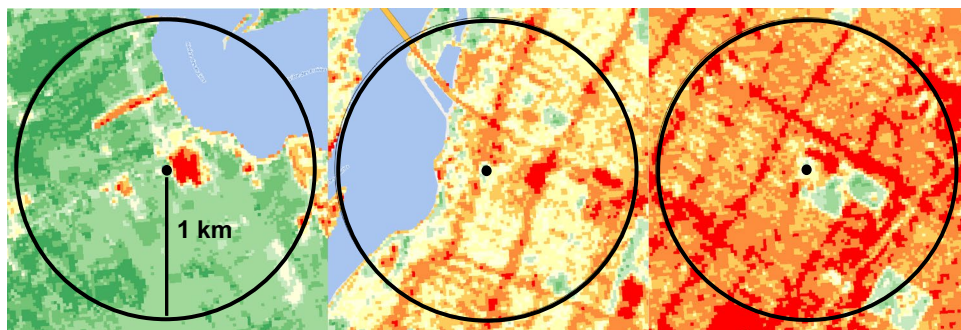


Fig. 2 Three examples of the heat island mapping of Boulfroy et al. (2013) and Institut national de santé publique du Québec (INSPQ) (2013) overlaid for our sites and their respective 1 km buffer zone. Sites depicted here have respectively the lowest (2.99%), median

(24.14%), and highest (68.22%) proportion of heat islands. The two highest levels of predicted heat islands are presented in dark orange (level 8) and red (level 9, highest intensity)

for traps of each trap colour during a given trapping period. These data were combined into detection histories, in which each detection history of a given species on a given site on a given date consisted of three digits: one for each trap colour (blue, white, or yellow). Because individuals were removed for identification, this may have modified species richness patterns on subsequent visits to the same sites. To account for this possibility, we estimated species richness separately for each site at each period, describing the potential change of species richness across the season. Furthermore, visits to the sites each year spanned six months during which species richness patterns may also have varied. For these reasons, we considered visits to a given site as spatial replicates (i.e., time for space substitution, *sensu* Kéry and Royle (2016)), yielding 312 site-date combinations. Nonetheless, we included a site random effect in our analysis to account for potential correlations among observations across visits.

We estimated species richness λ_i on site j on date k with a log link function for the linear predictor:

$$\log(\lambda_{jk}) = \beta_0 + \beta_{\log\text{HeatIslandCover}_j} \times \log\text{HeatIslandCover}_j + \beta_{\log\text{SiteArea}_j} \times \log\text{SiteArea}_j$$

where β corresponds to parameter estimates on the log scale, and where $\log\text{HeatIslandCover}_j$ and $\log\text{SiteArea}_j$ were the log of the explanatory variables for a given site j on the number of species. In the data augmentation component of the model, the true occurrence z_{ijk} of each species i at site j on date k was drawn from a Bernoulli distribution with probability φ , $z_{ijk} \sim \text{Bernoulli}(\varphi)$, and where φ is a derived parameter computed as $\varphi = \frac{\sum_{jk}^M \lambda_{jk}}{M}$. Here, M was set to 2000, representing the potential sum of species across the 312 site-date combinations (Kéry and Royle 2016).

Detection probability of species i on site j on sampling date k and trap colour l was estimated with a logistic link function based on the number of traps of colour l (sampling effort), Julian date, Julian date squared, and pan trap colour (blue, yellow, or white; blue being the reference level):

$$\log(p_{ijkl}/(1 - p_{ijkl})) = \alpha_0 + \alpha_{\text{Effort}} \times \text{Effort}_{jk} + \alpha_{\text{Jday}} \times \text{Jday}_{jk} + \alpha_{\text{Jday}^2} \times \text{Jday}^2_{jk} + \alpha_{\text{White}} \times \text{White}_{jkl} + \alpha_{\text{Yellow}} \times \text{Yellow}_{jkl} + \eta_{jk}$$

where α denotes parameter estimates of detection probability on the logistic scale and where η_{jk} is a random effect associated with site j on sampling date k with variance $\sigma^2_{\text{site}} (\eta_j \sim N(0, \sigma^2_{\text{site}}))$.

We estimated the parameters of the model by adopting a Bayesian approach based on Markov chain Monte Carlo (MCMC) using five chains (Gelman et al. 2014). Each chain consisted of 50,000 iterations, using the first 30,000 iterations as burn-in and a thinning rate of 5. We used vague prior distributions for all parameters. For instance, we considered

normal priors with $N(\mu=0, \sigma^2=100)$ for the fixed effect β parameters, and a uniform prior $U(0, 50)$ for the standard deviation parameter of random intercepts. Model structure and priors are presented in Table S1. This model was implemented in JAGS 4.3.0 within R 4.1.2 with the jagsUI and coda packages (Kellner 2017; Lunn et al. 2013; Plummer et al. 2006; R Core Team 2021). We evaluated convergence based on trace plots, posterior density plots, and the Brooks-Gelman-Rubin statistic. Model fit was assessed with a posterior predictive check comparing the observed number of species detected on site j on date k in trap colour l against the number of expected species detected with a Pearson chi-square (Kéry and Royle 2016). We reported means and 95% Bayesian credible intervals for each parameter of interest, where intervals excluding 0 indicated that the effect of a variable is different from 0.

Functional group occupancy

We investigated occupancy patterns of three of the four syrphid fly functional groups defined above (insectivores, terrestrial saprophagous, and aquatic saprophagous). The phytophagous functional group was excluded due to very sparse data. We combined the detection data of the species of each functional group for each sampling visit on the 25 sites sampled. Here, we translated capture data for each functional group into detected (1) or not detected (0) for traps of each trap colour during a given trapping period. Much like in the species richness analysis above, data were combined into detection histories.

Detection histories form the raw data to estimate the occupancy and detection probability of a given species or group with the classic occupancy model (MacKenzie et al. 2002). This occupancy model includes two main parameters: one describes the biological process of interest, which is the probability of occupancy of the species (ψ), whereas the other describes the observation process (p). To account for potential heterogeneity in either occupancy or detection probability, covariates can be included on these parameters using a logit link function analogous to logistic regression. We modified the approach to our syrphid fly functional group data to test our hypotheses on site surface area and heat island cover within 1 km. Specifically, we estimated occupancy and detection probability of each functional group in the same hierarchical model.

We modeled the occupancy of functional group i on site j on a given sampling date k :

$$\log(\psi_{ijk}/(1 - \psi_{ijk})) = \beta_{0,i} + \beta_{\log\text{HeatIslandCover}_j} \times \log\text{HeatIslandCover}_j + \beta_{\log\text{SiteArea}_j} \times \log\text{SiteArea}_j$$

where $\log\text{HeatIslandCover}_j$ and $\log\text{SiteArea}_j$ are the same explanatory variables as for the species richness analysis

above. This parameterisation of the model produced three unique β estimates for each functional group.

We modeled the detection probability of functional group i on site j on date k in trap colour l as a function of the number of traps of a given colour (sampling effort), Julian date, Julian date squared, and pan trap colour (blue, yellow, or white; blue being the reference level):

$$\log(p_{ijkl}/(1-p_{ijkl})) = \alpha_{0,i} + \alpha_{\text{Effort},i} \times \text{Effort}_{jkl} + \alpha_{\text{Jday},i} \times \text{Jday}_{jkl} + \alpha_{\text{Jday2},i} \times \text{Jday2}_{jkl} + \alpha_{\text{White},i} \times \text{White}_{jkl} + \alpha_{\text{Yellow},i} \times \text{Yellow}_{jkl} + \eta_j$$

where each α_i (i.e., intercept, effort, Jday, Jday2, White, and Yellow) denotes the effect of a given variable on the detection probability of functional group i , and where η_j corresponds to the random effect of detection probability associated with site j , normally distributed with hyperparameters mean μ_{site} and variance σ^2_{site} ($\eta_j \sim N(0, \sigma^2_{\text{site}})$).

Our hierarchical model included fixed effects for each functional group and random intercepts, similar to a generalized linear mixed model (Gelman and Hill 2007; Zuur et al. 2009). Parameters were estimated with MCMC and five chains. Each chain was run with 1,000,000 iterations, using 750,000 iterations as burn-in, and a thinning rate of 25. We used normal priors $N(\mu=0, \sigma^2=10)$ for the predictors and uniform priors $U(0, 20)$ for the standard deviation of the site random effects. The complete code using JAGS syntax is presented in Table S2. We used trace plots, posterior density plots, and the Brooks-Gelman-Rubin statistic to assess chain convergence. Model fit was assessed with posterior predictive checks based on a Pearson chi-square aggregated over rows and columns (Kéry and Royle 2016). In addition, we computed the area under the receiver operating characteristic (ROC) curve as a measure of predictive ability (Zipkin et al. 2012). This metric can be used to assess to what extent the discriminatory ability of the model is better than random (i.e., >0.5).

Results

A total of 1,791 specimens were captured in the 2012 and 2013 sampling seasons, representing 48 species or genera (Table 1). Cemeteries harboured 77% of the species captured in the study area, whereas gardens and parks included 46% and 42% of all species, respectively. Eight species were captured on all site types. The number of specimens captured on each site type varied greatly. Eighty-three percent of the specimens were collected in cemeteries, whereas gardens (11%) and parks (6%) yielded fewer specimens. Species or genera were characterised according to the larval functional groups. Four functional groups were represented in the capture data (16 insectivore species, 14 terrestrial saprophagous species, 16 aquatic saprophagous species, and

2 phytophagous species), although most specimens captured in our study were insectivores (92%). Aquatic saprophagous, terrestrial saprophagous, and phytophagous species corresponded to only 4%, 3%, and 1% of the overall specimens. *Toxomerus marginatus* (Say, 1823) and *Toxomerus geminatus* (Say, 1823), two insectivore species, were the most abundant species and represented 66.9% and 20.7% of all specimens captured, respectively. The species accumulation curve appeared to stabilize after 300 samples, suggesting that the sampling effort was adequate to detect the species present (Fig. 3).

Species richness

Trace plots and other diagnostics for the species richness model suggested that chains mixed well and were of sufficient length. The posterior predictive check suggested good model fit ($\chi^2=932.87$, $P=0.48$). Species richness increased with site surface area (Table S3, Fig. 4) but did not vary with the cover of heat islands within a 1,000 m radius around sites (Table S3). Species detection probability did not vary with sampling effort (Table S3). Detection probability followed a quadratic trend with Julian date, reaching a peak in mid-season (Table S3, Fig. 5). Finally, detection probability of species with blue pan traps was lower than with yellow traps but did not differ from that of white traps (Table S3, Fig. 5).

Functional group occupancy

The occupancy analysis focused on three functional groups: insectivores, terrestrial saprophagous, and aquatic saprophagous. The phytophagous functional group was excluded due to very sparse data. Trace plots and other diagnostics for the occupancy model on functional groups suggested that chains mixed well for all parameters and that chains were sufficiently long. There were no indications that the model lacked fit, based on the posterior predictive check on either rows or columns ($\chi^2_{\text{rows}}=1096.87$, $P=0.20$; $\chi^2_{\text{columns}}=7.26$, $P=0.49$). The area under the ROC curve was 0.64 (95% CRI: [0.54, 0.75]), suggesting an intermediate model performance.

Our functional group analysis revealed that the occupancy of the three groups did not vary with site surface area. However, the groups responded differently to heat island cover (Table S4). The occupancy of syrphid flies of the terrestrial saprophagous group decreased with increasing heat island cover (Table S4, Fig. 6). In contrast, the occupancy of the insectivore and the aquatic saprophagous groups did not vary with the cover of heat islands within 1,000 m (Table S4).

The detection probability of the three functional groups increased with sampling effort, although the relationship was weaker for saprophagous syrphids (Table S4, Fig. 7a, c, e). Detection probability of insectivores varied quadratically

Table 1 Syrphid fly species and number of specimens captured in 2012 and 2013 in Montreal's urban gardens, parks, and cemeteries

Species	Total number of individuals	Cemeteries	Gardens	Parks	Functional group
<i>Allograpta obliqua</i> (Say, 1823)	3	3	0	0	I
<i>Anasimyia bilinearis</i> (Williston 1887)	2	0	0	2	AS
<i>Brachyopa vacua</i> Osten Sacken, 1875	1	1	0	0	TS
<i>Chalcosyrphus nemorum</i> (Fabricius, 1805)	1	1	0	0	TS
<i>Chrysotoxum plumeum</i> Johnson, 1924	3	3	0	0	I
<i>Didea fuscipes</i> Loew, 1863	1	1	0	0	I
<i>Epistrophe nitidicollis</i> (Meigen, 1822)	10	8	0	2	I
<i>Eristalis anthophorina</i> (Fallen, 1817)	1	0	1	0	AS
<i>Eristalis arbustorum</i> (Linnaeus, 1758)	19	2	16	1	AS
<i>Eristalis dimidiata</i> (Wiedemann, 1830)	5	3	2	0	AS
<i>Eristalis tenax</i> (Linnaeus, 1758)	3	3	0	0	AS
<i>Eristalis transversa</i> (Wiedemann, 1830)	5	3	2	0	AS
<i>Eumerus</i> sp.	7	5	2	0	P
<i>Eupeodes americanus</i> (Wiedemann, 1830)	19	16	2	1	I
<i>Eurimyia stipata</i> (Walker 1849)	1	0	0	1	AS
<i>Helophilus fasciatus</i> Walker, 1849	10	8	2	0	AS
<i>Helophilus latifrons</i> Loew, 1863	4	3	1	0	AS
<i>Mallota bautias</i> (Walker, 1849)	9	7	0	2	AS
<i>Merodon equestris</i> (Fabricius, 1794)	7	1	5	1	P
<i>Orhonevra pictipennis</i> (Loew, 1863)	1	0	1	0	AS
<i>Parhelophilus laetus</i> (Loew, 1963)	2	0	1	1	AS
<i>Pelecinobaccha costata</i> (Say 1829)	1	1	0	0	I
<i>Pipiza</i> sp.	2	1	0	1	I
<i>Platycheirus</i> sp.	1	1	0	0	I
<i>Rhingia nasica</i> Say, 1823	1	0	0	1	TS
<i>Sericomyia bifasciata</i> Williston, 1887	1	1	0	0	AS
<i>Sericomyia chrysotoxoides</i> Macquart, 1842	3	3	0	0	AS
<i>Sericomyia militaris</i> Walker, 1849	5	5	0	0	AS
<i>Somula decora</i> Macquart, 1847	4	2	0	2	TS
<i>Sphaerophoria</i> sp.	25	24	1	0	I
<i>Spilomyia longicornis</i> Loew, 1872	5	2	1	2	TS
<i>Spilomyia sayi</i> (Goot, 1964)	5	1	1	3	TS
<i>Syrpitta pipiens</i> (Linnaeus, 1758)	17	3	14	0	TS
<i>Syrphus attenuatus</i> Hine, 1922	2	1	0	1	I
<i>Syrphus knabi</i> Shannon, 1916	1	0	1	0	I
<i>Syrphus rectus</i> Osten Sacken, 1875	1	0	1	0	I
<i>Syrphus ribesii</i> Linnaeus	1	0	1	0	I
<i>Syrphus torvus</i> Osten Sacken, 1875	1	1	0	0	I
<i>Temnostoma balyras</i> (Walker 1849)	1	0	0	1	TS
<i>Temnostoma barberi</i> Curran 1939	1	1	0	0	TS
<i>Temnostoma excentrica</i> (Harris, 1841)	1	1	0	0	TS
<i>Toxomerus geminatus</i> (Say, 1823)	370	285	31	54	I
<i>Toxomerus marginatus</i> (Say, 1823)	1,198	1,068	108	22	I
<i>Tropidia quadrata</i> (Say, 1824)	9	0	1	8	AS
<i>Xylota annulifera</i> Bigot, 1884	2	2	0	0	TS
<i>Xylota quadrimaculata</i> Loew, 1866	5	3	1	1	TS
<i>Xylota segnis</i> (Linnaeus, 1758)	13	12	0	1	TS
<i>Xylota subfasciata</i> Loew, 1866	1	1	0	0	TS
Total number of individuals	1,791	1,487	196	108	

Trapping occurred over 13 periods of sampling, using blue, white, and yellow coloured pan traps. Functional groups consist of insectivores (I), terrestrial saprophagous (TS), aquatic saprophagous (AS), or phytophagous (P) species

Fig. 3 Species accumulation curve of syrphid flies captured in the Montreal metropolitan area, QC, Canada in 2012 and 2013. Error bars denote 95% confidence intervals based on 999 permutations

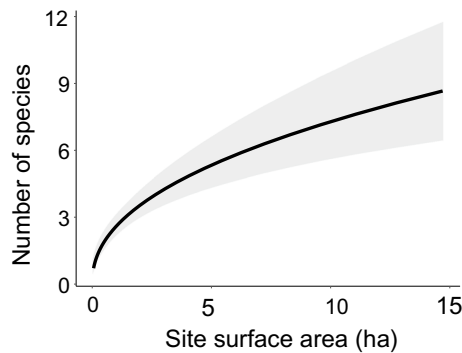
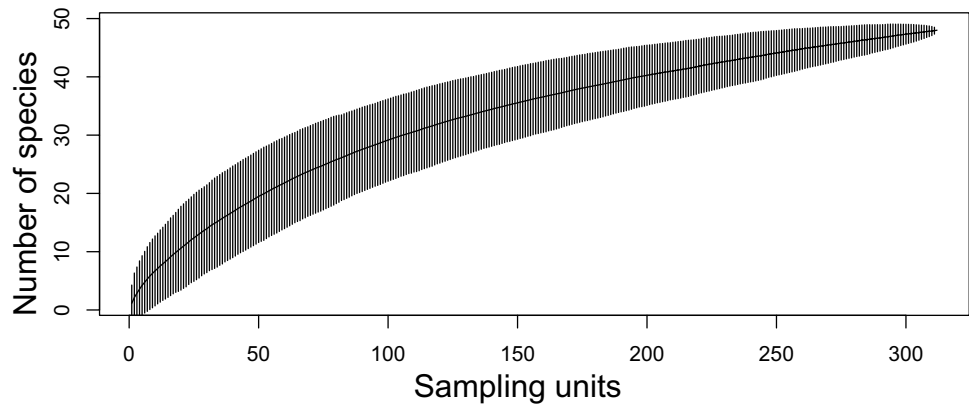


Fig. 4 Species richness of syrphids estimated across site surface area from captures in the Montreal metropolitan area, QC, Canada in 2012 and 2013. Error bars denote 95% Bayesian credible intervals around estimates

with the Julian date, with the highest detection probability in mid-season, whereas detection probability of the two other groups increased throughout the season (Table S4, Fig. 7b, d, f). For the three groups, detection probability was highest with yellow pan traps (Table S4, Fig. 7g). The effectiveness of white traps was intermediate for aquatic saprophagous species but did not vary relative to blue traps for the other two functional groups (Table S4, Fig. 7g).

Discussion

We present results from more than 300,000 trap-hours. As suggested by the species accumulation curve, this effort yielded a representative sample of the syrphid fly community of the Montreal metropolitan area in Quebec, Canada. To our knowledge, this study provides the first inventory of syrphid flies conducted in an urban environment in Canada and therefore offers an unprecedented portrait of their populations. We captured 1,791 specimens of syrphid flies representing 48 species in parks, cemeteries, and gardens. The number of individuals and species captured is

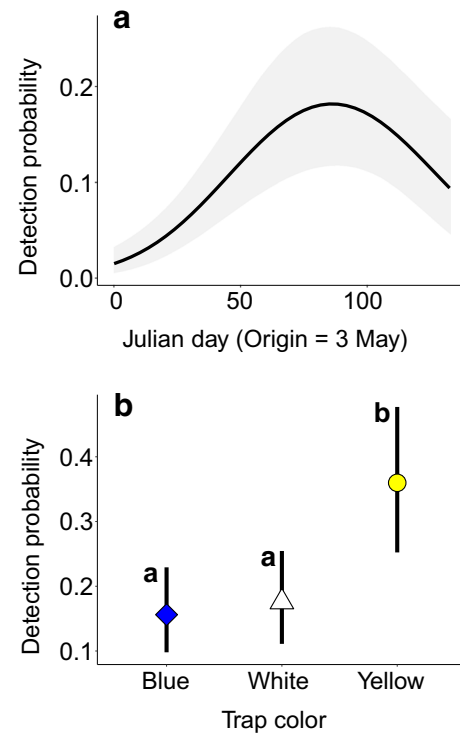


Fig. 5 Detection probability of syrphid fly species across Julian day (a) and pan trap colours (b) based on syrphid flies captured in the Montreal metropolitan area, QC, Canada in 2012 and 2013. Error bars denote 95% Bayesian credible intervals. For pan trap colour, letters indicate differences among colours

greater than reported in other studies conducted in Quebec in lowbush blueberry (430 specimens of 28 species) or cranberry crops (461 specimens of 33 species) (Gervais et al. 2018; Moisan-DeSerres et al. 2015), thus extending our knowledge on the syrphid flies of Quebec. Regarding common species, other studies also found *T. marginatus* to be the most abundant syrphid species in Quebec (Gervais et al. 2018; Moisan-DeSerres et al. 2015). Furthermore, individuals with aphidophagous larvae constituted large proportions of the syrphid fly communities in previous

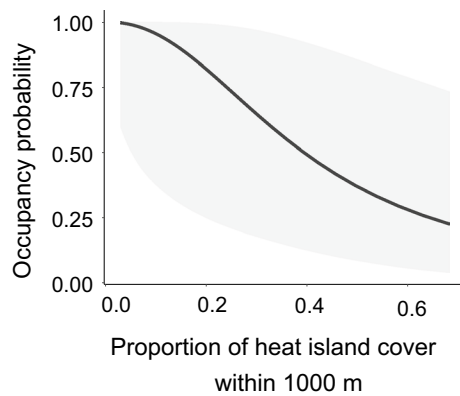


Fig. 6 Occupancy probability of syrphids of the terrestrial saprophagous functional group across heat island cover based on syrphid flies captured in the Montreal metropolitan area, QC, Canada in 2012 and 2013. Error bars denote 95% Bayesian credible intervals

studies (Burgio and Sommaggio 2007; Meyer et al. 2009; Moquet et al. 2018).

We found a positive relationship between syrphid fly species richness and the site surface area. This result is consistent with previous observations that syrphid fly species richness increases with habitat heterogeneity parameters such as area of habitat and landscape diversity (Burgio and Sommaggio 2007; Herrault et al. 2016; Jauker et al. 2019; Meyer et al. 2009; Ouin et al. 2006). The life cycle of syrphid flies is complex and requires different resources at both larval and adult stages (Meyer et al. 2009; Sommaggio 1999). Larger sites are usually more heterogeneous (Ouin et al. 2006) and harbor more diversified and abundant resources and can thus satisfy more syrphid fly species at all stages in their life cycle. Larger sites also hold more diverse and abundant nectar and pollen sources. Kleijn and van Langevelde (2006) showed that the amount of habitat at the landscape level only influences syrphid fly richness when the floral resources are abundant at the local scale.

Surprisingly, our results indicate that urbanisation only marginally affects syrphid flies at the landscape scale. Indeed, we only detected a negative impact of urbanisation on the terrestrial saprophagous syrphids as the occupancy of this group increased with decreasing cover of heat islands within 1,000 m (Fig. 6). Similarly, Rocha et al. (2018) observed no impacts of impervious surface cover, a parameter closely related to heat islands (Hart and Sailor 2009), on the occurrence of syrphid flies. Interestingly, prior studies showed that urbanisation mostly reduces the abundance and diversity of dipterans, including syrphid flies (Deguines et al. 2012, 2016; Persson et al. 2020; Verboven et al. 2014). However, other studies report contradictory results for urban sites. For example, Baldock et al. (2015) found that urban sites in the United Kingdom harboured less abundant syrphid fly populations

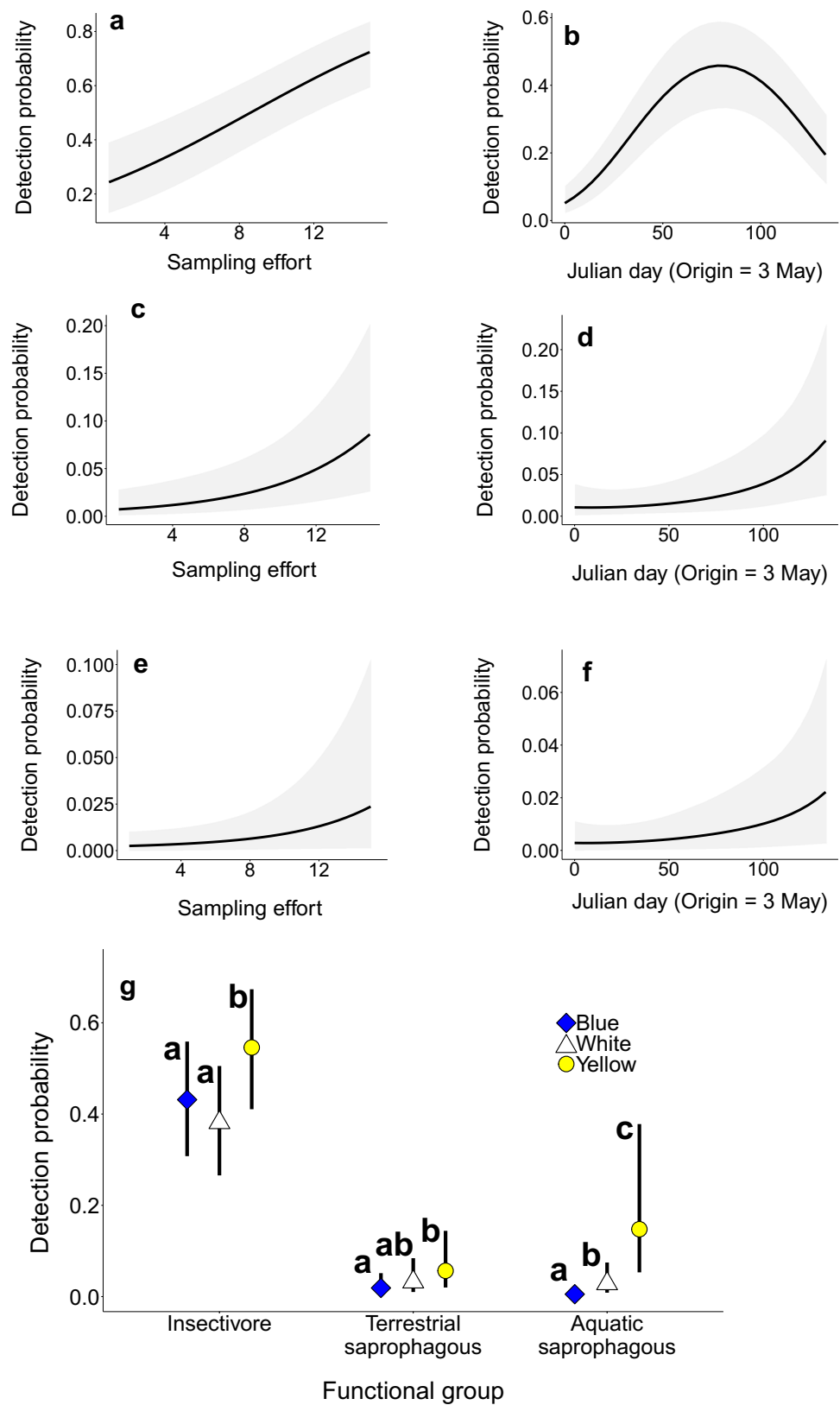
compared to natural or rural sites but had similarly rich communities. Bates et al. (2011) found that urban and suburban sites harboured fewer species than farmlands, but not fewer individuals. Abundance was also lower on urban sites in Switzerland, but diversity was not (Luder et al. 2018). In our study, we did not compare the abundance and diversity of syrphids in urban areas to those of natural or rural sites. Nonetheless, heat island cover varied from 2.99 to 68.22%, representing a gradient for the study of impacts of urbanisation on syrphid communities.

Estimating detection probability was crucial in our study to estimate meaningful parameters of species richness and occupancy. As in previous studies in which yellow traps were most effective in attracting syrphid flies (Laubertie et al. 2006), our probability of detecting species was greatest in yellow bowls (Fig. 7). Syrphid flies use visual cues when foraging and have a colour preference for white and yellow flowers (Haslett 1989; Klecka et al. 2018; Sutherland et al. 1999). Such observations may explain the high attractivity of our yellow bowls. However, our results contrast with those of other studies that found blue traps to be highly attractive to syrphid flies (Campbell and Hanula 2007; Chen et al. 2004). Finally, the positive relationship between detection probability and sampling effort is straightforward, whereas differences in detection probability with the linear or quadratic effects of Julian date reflect phenological differences across functional groups throughout the season. Species richness and the occupancy of insectivores peaked by mid-summer, while aquatic syrphid flies reached a peak at the end of the season.

Conclusions

In conclusion, we showed that urban green spaces can harbour species-rich and abundant syrphid fly communities. We investigated the impact of heat islands on syrphid flies in an urban landscape. We only found a negative relationship between terrestrial saprophagous syrphid flies and the cover of urban heat islands, which suggests that the urbanisation threshold at which changes in land use become highly detrimental was not reached for the two other functional groups. Our results will be useful in developing land management and conservation measures to promote syrphid fly communities. Beneficial measures implemented in highly urbanised sectors, such as sowing or planting floral stripes or patches (Carreck and Williams 2002), should be targeted at vulnerable groups such as saprophagous syrphid flies. Land management practitioners should also ensure that floral resources are available continuously throughout the season, and that larval requirements can be met, for example, through microhabitat elements such as decaying wood debris and standing water (Cole et al. 2022). Finally, our results highlight that

Fig. 7 Detection probability of syrphid fly functional groups across sampling effort (insectivores (**a**), terrestrial saprophagous (**c**), and aquatic saprophagous (**e**)), Julian day (insectivores (**b**), terrestrial saprophagous (**d**), and aquatic saprophagous (**f**)), and pan trap colour (**g**) based on syrphid flies captured in the Montreal metropolitan area, QC, Canada in 2012 and 2013. Letters indicate differences among pan trap colors for a given functional group. Error bars denote 95% Bayesian credible intervals



conservation measures should be adapted to syrphid functional groups and that appropriate measures may depend on larval requirements.

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Author contributions FMc: conceptualization, methodology, investigation, writing—original draft; ÉN: conceptualization, methodology, investigation, writing—review & editing; AG: investigation, writing—review & editing; MJM: formal analysis, writing—review & editing, supervision; VF: conceptualization, methodology, resources, writing—review & editing, supervision, funding acquisition.

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Declarations

Conflict of interest The authors have no conflicts of interest to disclose.

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